

# 2024-10-28-ApplsLinProg

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December 16, 2024

## Slide 1

### Content

Production Planning  
[M&D] p.16    2.3 Ice Cream All Year Round

### Description

This slide introduces a production planning problem for an ice cream manufacturer. The title "Production Planning" is underlined. A reference to a textbook or other source ([M&D] p.16) is given, along with the section number (2.3) and title "Ice Cream All Year Round". The main content is a bar chart illustrating monthly ice cream demands ( $d_i$ , where  $i$  represents the month from 1 to 12) in tons. The chart shows the predicted monthly sales of ice cream for the next year, providing data for a linear programming optimization problem. The context of the course (Linear Programming and Optimization Techniques) indicates that this data will be used to formulate a linear program to optimize the ice cream production plan, likely aiming to minimize costs or maximize profits while meeting the predicted demand.

### Figures

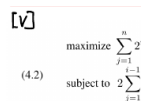

$$\begin{aligned} & \text{[v]} \\ & \text{(4.2)} \quad \begin{aligned} & \text{maximize} \quad \sum_{j=1}^n 2 \\ & \text{subject to} \quad 2 \sum_{j=1}^{i-1} \end{aligned} \end{aligned}$$

Figure 1: A bar chart showing monthly demands ( $d_i$ ,  $i = 1, \dots, 12$ ) for ice cream in tons. The x-axis represents the months (Jan-Dec), and the y-axis represents the sales in tons.

## Slide 2

### Content

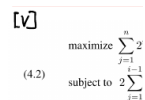
Production Planning

[M&D] p.16    2.3 Ice Cream All Year Round

### Description

This slide presents a production planning problem for Icicle World Ltd., an ice cream manufacturer. The title "Production Planning" is underlined. A reference to a textbook or other source ([M&D] p.16) is given, along with the section number (2.3) and title "Ice Cream All Year Round". The main content is a bar chart illustrating the monthly ice cream demand ( $d_i$ , where  $i$  represents the month from 1 to 12) in tons. The chart shows predicted monthly sales for the next year. A horizontal red line indicates the average monthly demand. The context of the course (Linear Programming and Optimization Techniques) indicates that this data will be used to formulate a linear program to optimize the ice cream production plan, likely aiming to minimize costs or maximize profits while meeting the predicted demand. The text mentions that the prediction is based on historical data, extensive surveys, and bird observations.

### Figures



$$(4.2) \quad \begin{array}{ll} \text{maximize} & \sum_{j=1}^n x_j \\ \text{subject to} & 2 \sum_{j=1}^{i-1} x_j \end{array}$$

Figure 2: A bar chart showing monthly demands ( $d_i$ ,  $i = 1, \dots, 12$ ) for ice cream in tons. The x-axis represents the months (Jan-Dec), and the y-axis represents the sales in tons. A horizontal line indicates the average demand.

## Slide 3

### Content

Production Planning

[M&D] p.16

- (1)  $x_i$  – production at month  $i$
- (2)  $s_i$  – surplus at end of month  $i$
- (3)  $x_i + s_{i-1} = d_i + s_i$

$$\min \sum_{i=1}^{12} 50|x_i - x_{i-1}| + \sum_{i=1}^{12} 20s_i$$

### Description

This slide formulates a linear programming model for a production planning problem. The title "Production Planning" is underlined. A reference to a textbook or other source ([M&D] p.16) is given. The model involves: (1)  $x_i$  representing the production level in month  $i$ ; (2)  $s_i$  representing the surplus inventory at the end of month  $i$ ; and (3) a constraint  $x_i + s_{i-1} = d_i + s_i$ , where  $d_i$  is the demand in month  $i$ , ensuring that production plus beginning inventory equals demand plus ending inventory. The objective function is to minimize the

total cost, which consists of two parts:  $\sum_{i=1}^{12} 50|x_i - x_{i-1}|$  representing the cost of changing production levels

between consecutive months, and  $\sum_{i=1}^{12} 20s_i$  representing the cost of storing surplus inventory. The context

of the course (Linear Programming and Optimization Techniques) indicates that this model will be solved using linear programming techniques. The absolute value in the cost of changing production levels suggests that this model might require reformulation to be solved using standard linear programming methods.

### Figures

Worst Case Ex  
[v]

$$(4.2) \quad \begin{array}{ll} \text{maximize} & \sum_{j=1}^n 2 \\ \text{subject to} & 2 \sum_{j=1}^{i-1} \end{array}$$

Figure 3: No figure present.